I've seen miracles in every way: On the Space Complexity of Miracle Sort

Jade Harley

SIGBOVIK 2026

Abstract

Miracle Sort is a sorting algorithm in which an array is sorted by miracles. A widely circulated image [9] (Figure 1) claims the algorithm has time complexity $O(\infty)$ and space complexity $O(1)$. We found the reasoning for each of these claims [2] to be unsatisfying and set out to solve the problem. We present a corrected formulation of Miracle Sort known as "Motherfuckin' Miracle Sort"[1] and prove that any correct implementation requires $O(\log N)$ bits of auxiliary space using multiset hashing. We leave the questions of how magnets work and affect our algorithm as an exercise for the cosmically irradiated reader.

1 Water, Fire, Air and Dirt

Sorting is among the most well-studied problems in computer science. Algorithms such as Quicksort [6], Bogosort [5], Sleep Sort [1], and Stalin Sort [8] have been studied heavily by undergraduate algorithms students who probably just have ChatGPT analyze them nowadays. Less attention has been paid to algorithms in which Jesus[2] takes the wheel.

Miracle Sort occupies a unique position in the taxonomy of sorting algorithms in that it doesn't do anything. The algorithm stores an array A of N integers drawn from $[0, k]$ (where k is probably something like $8^j - 1$ where $j \in \mathbb{N}^+$) and waits for a miracle (e.g. random bit flips induced by cosmic rays) to transform A into a sorted permutation of itself. We note that while the original algorithm simply states that a miracle occurs, our analysis is the same regardless of whether that miracle is random bitflips or divine intervention. We describe some of the problems of random bitflips in Section 2.

The algorithm's existence was popularized by the image shown in Figure 1, which asserts a time complexity of $O(\infty)$ and a space complexity of $O(1)$. While the former claim is difficult to dispute, we contend that the latter is **incorrect**, due to its overreliance on faith-based arguments [3].

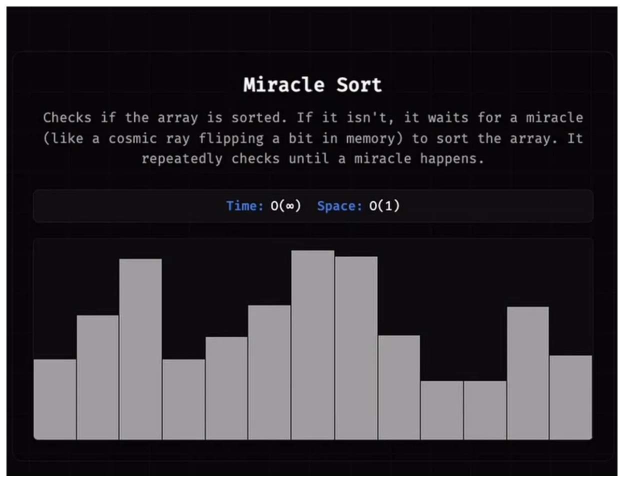

Figure 1: The widely circulated claim regarding Miracle Sort's complexity. The time complexity is accepted without comment. The space complexity is the subject of this paper.

2 I Don't Wanna Talk to a Scientist

A sorting algorithm must, at minimum, produce output that is (1) sorted in non-decreasing order and (2) a permutation of the input. The naïve formulation of Miracle Sort addresses only the first criterion.

Consider the input array $[0, 6, 3]$. Under random bit flipping, a possible intermediate state is $[0, 2, 3]$. This array is sorted, yet it does not contain the same elements as the original. A sorting algorithm that returns $[0, 2, 3]$ when given $[0, 6, 3]$ is not a sorting algorithm at all; it is, at best, an optimistic guess.

One might argue that if the miracle is truly divine rather than stochastic, then the sorted array

[1] After the Insane Clown Posse song *Miracles* [7], which serves as the theoretical foundation for this work.

[2] Or your mythological being of choice.

is *guaranteed* to be a permutation of the input, because God would not make such an elementary mistake. We reject this line of reasoning on empirical grounds [4]: extraordinary claims about sorting correctness require extraordinary proof, and "a miracle happened" does not constitute a formal verification. Even if we grant omniscience to the sorting agent, we, the humble verifiers, still need a way to check.

We therefore assert that Miracle Sort must verify not only that the array is sorted but that the resulting array is a permutation of the input. This verification demands auxiliary information, which in turn demands auxiliary space. The central question of this paper is: *how much space does Miracle Sort require?*

3 Pure Motherfuckin' Magic

One straightforward approach is to store a complete copy of the original array and, upon observing a sorted state, verify element-by-element membership via binary search. This requires $O(N)$ space and $O(N \log N)$ verification time. Given algorithms professors live purely for the joy of crushing the hopes and dreams of students who say things like this, let's dispel any thoughts about naiv̈e approaches.

A more elegant solution employs *multiset hashing*: a hash function h such that $h(A) = h(B)$ if and only if A and B are equal as multisets (i.e., they contain the same elements with the same multiplicities, regardless of ordering). We compute $H = h(A)$ once before the miracle begins and store only this hash. The modified algorithm is presented as Algorithm 1.

We assume, optimistically, that the hash value H is not itself corrupted by the same cosmic rays responsible for sorting the array. Figure 2 illustrates the required system architecture.

Algorithm 1 Motherfuckin' Miracle Sort

```
H ← hash(A)
while true do
    Wait for cosmic rays or divine intervention
    if is_sorted(A) and hash(A) = H then
        return A
    end if
end while
```

4 I've Seen It with My Own Eyes

We now derive the minimum size of a perfect multiset hash. A perfect multiset hash must assign a distinct value to every possible multiset of size N with elements drawn from $\{0, 1, \ldots, k\}$. The number of such multisets is given by the stars-and-bars formula:

$$\binom{N+k}{N} = \frac{(N+k)!}{N! \cdot k!}$$

Expanding as a product of k terms in the numerator:

$$\frac{(N+k)!}{N! \cdot k!} = \frac{(N+k)(N+k-1)\cdots(N+1)}{k!}$$

For large N relative to k, each of the k terms in the numerator is approximately N, giving:

$$\frac{(N+k)(N+k-1)\cdots(N+1)}{k!} \approx \frac{N^k}{k!}$$

The number of bits required to represent this many distinct hash values is:

$$\log_2 \frac{N^k}{k!} = k \log_2(N) - \log_2(k!)$$

Since k is fixed (determined by the element type), the term $\log_2(k!)$ is constant with respect to N. The space complexity of the hash is therefore $O(\log N)$ bits. This is information-theoretically optimal: no hash smaller than this can distinguish all possible multisets.

Theorem 1 *Assuming k is constant, the space complexity of a correct implementation of Miracle Sort is* $\boxed{O(\log N) \text{ bits}}$.

This is strictly better than the $O(N)$ bits required to store a full copy of the input, and strictly worse than the (unproven) $O(1)$ claimed in Figure 1.

5 This world is yours for you to explore

Many questions remain after this work. For example:

1. Fucking magnets, how do they work?

2. Can Miracle Sort be parallelized? If N computers each wait for a miracle, do miracles arrive N times faster, or does God have a rate limit?

3. Does Carrie Underwood know what a sorting algorithm is? [10]

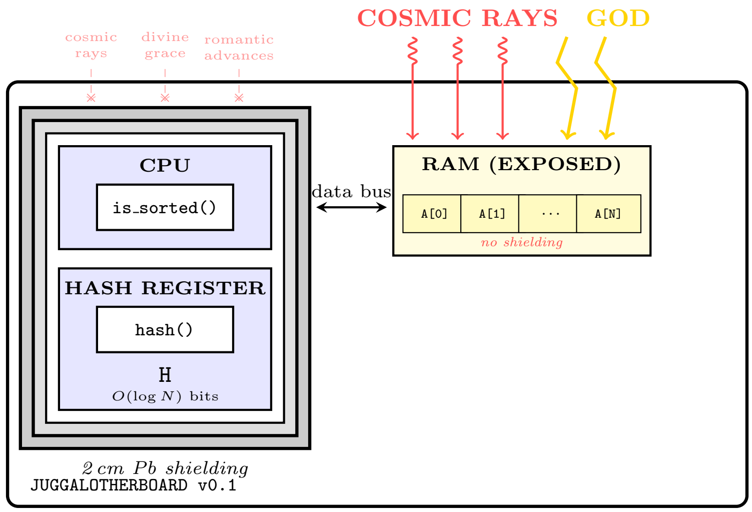

Figure 2: Required system architecture for Motherfuckin' Miracle Sort. The array resides in unshielded RAM, exposed to cosmic rays and/or divine intervention. The hash H is stored in a lead-lined vault. We recommend at minimum 2 cm of Pb shielding, or sufficient blasphemy to repel divine intervention.

References

[1] Anonymous. Sleep sort, 2011. Posted on 4chan's /prog/ board. Sorts by sleeping for a duration proportional to each element's value. Technically $O(max(A))$.

[2] Anonymous. Source: Just trust me bro, 2024.

[3] Thomas Aquinas. *Summa Theologica*. Self-published (posthumously), 1274. Contains no analysis of sorting algorithms, which we consider a significant oversight.

[4] Richard Dawkins. *The God Delusion*. Bantam Press, 2006.

[5] Folklore. Bogosort, Immemorial. The second-worst sorting algorithm. Miracle Sort is the first.

[6] C. A. R. Hoare. Quicksort. *The Computer Journal*, 5(1):10–16, 1962. A sorting algorithm that actually works.

[7] Insane Clown Posse. Miracles, 2009. From the album *Bang! Pow! Boom!*. A foundational text in the philosophy of science.

[8] mathew. I came up with a single pass O(n) sort algorithm i call StalinSort. `https://mastodon.social/@mathew/100958177234287431`, 2018. Posted on Mastodon. Elements that are not in sorted order are eliminated. $O(n)$ time, $O(1)$ space, $O(n)$ casualties.

[9] Mehul Mohan. Found my new favourite sorting algorithm. `https://x.com/mehulmpt/status/2035818992007512561`, 2025. Posted on Twitter/X. The post that started it all.

[10] Carrie Underwood. Jesus, take the wheel, 2005. An early proposal for faith-based autonomous systems. Does not address sorting.